

INTRODUCTION TO LUBRICANTS



UNIT-2

LUBRICANTS

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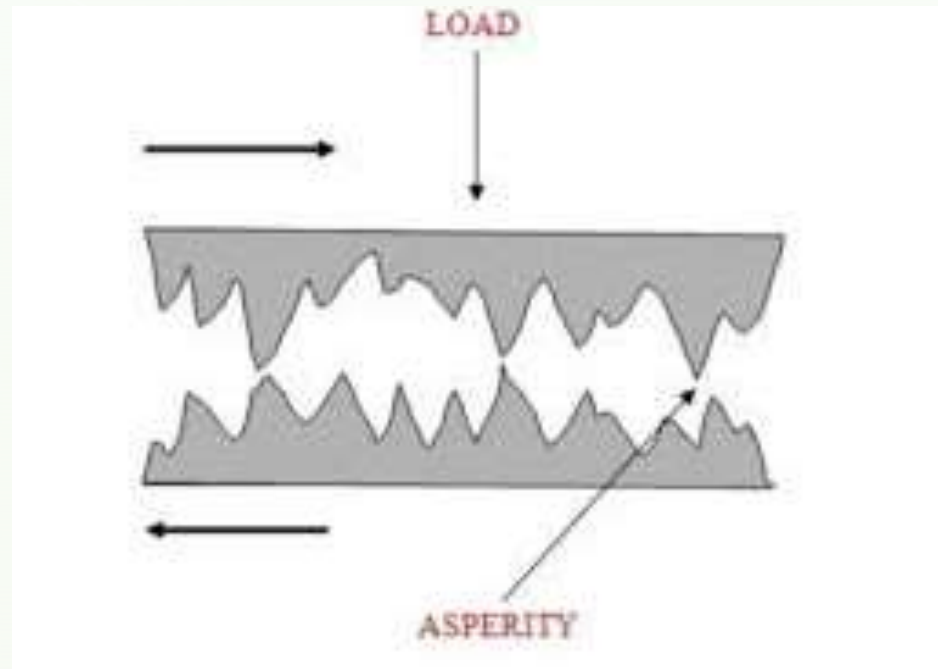
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Friction

- Friction is the **resisting force** that occurs when two surfaces move across each other.
- It acts **opposite to the direction of motion**.
- Caused by **microscopic irregularities** and **intermolecular interactions** between surfaces.
- Can be **static** (before motion starts) or **kinetic/dynamic** (during motion)

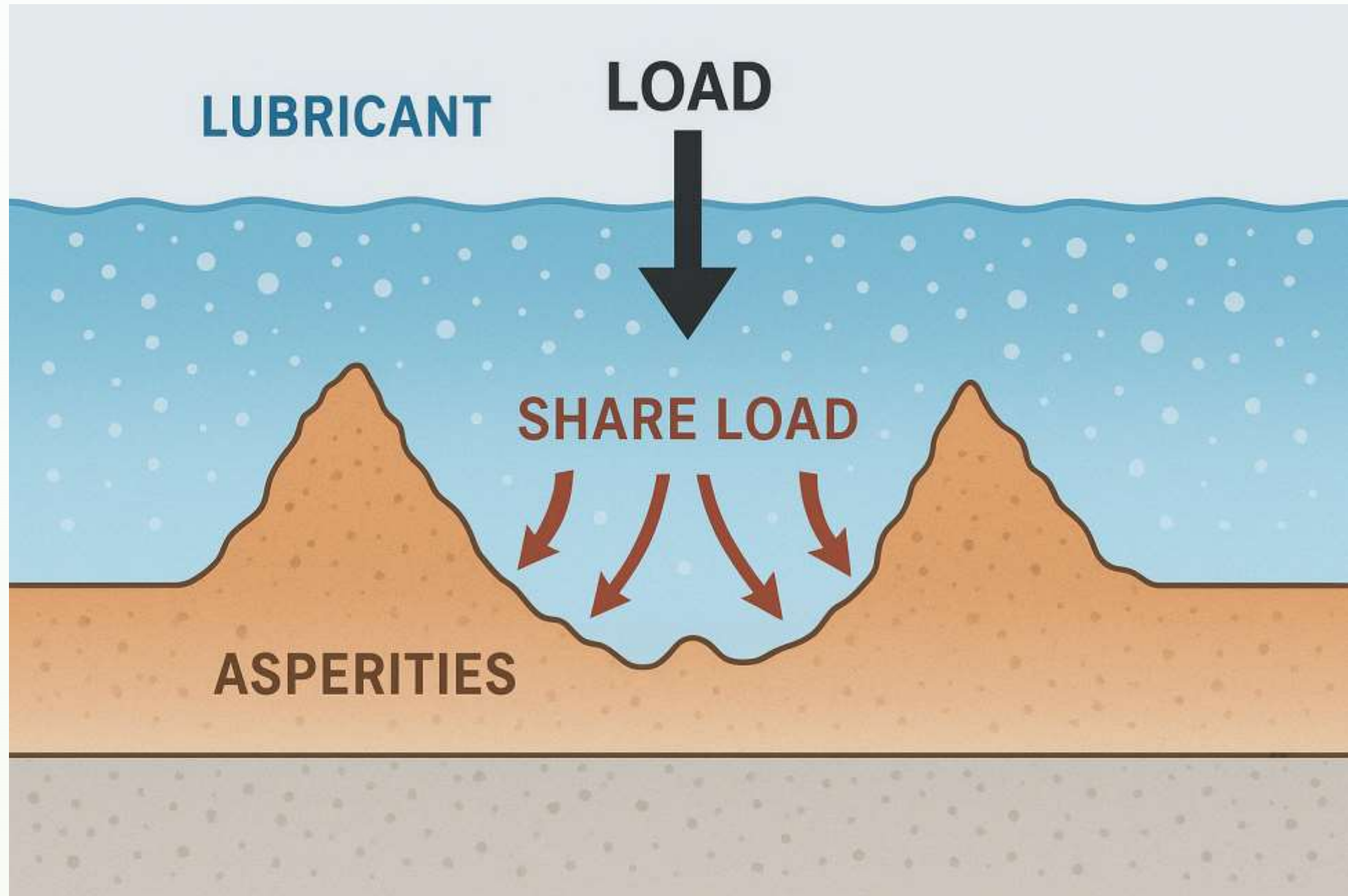


Frictional Heat

- The **energy lost due to friction** is converted into **heat energy**.
- Generated by **relative motion** and **surface interactions**.
- Excessive frictional heat can lead to **temperature rise and material degradation**,
- Important in **machinery**, **braking systems**, and **manufacturing processes**.

Lubricant

- ✓ A **substance introduced between two moving surfaces** to reduce friction and wear.
- ✓ Can be **liquid**, **solid**, **semi-solid**, or **gaseous** in form.
- ✓ Primary functions: **minimize friction, reduce wear, dissipate heat, and prevent corrosion**.
- ✓ e.g. **mineral oils, synthetic oils, greases, and solid lubricants** (e.g., graphite, MoS_2).



Lubrication

- ✓ The **process or technique of applying a lubricant** between contact surfaces.
- ✓ Aims to **reduce direct metal-to-metal contact** during motion.
- ✓ Ensures **smooth operation, increased efficiency**, and **longer component life**.
- ✓ Involves formation of a **lubricating film** that supports the load and separates surfaces.



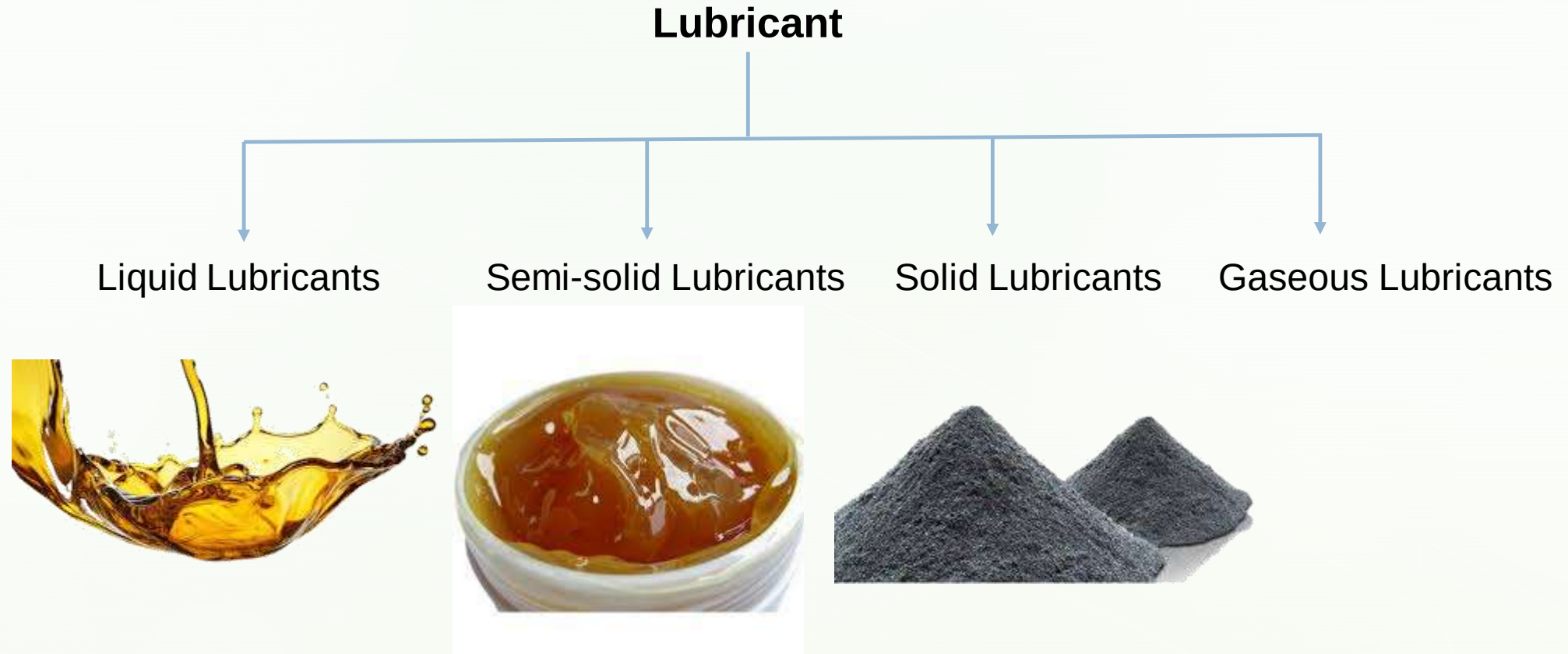
Function of lubricants

- Reduce friction
- Reduce deformation
- Reduce wear & tear
- Works as a coolant
- Works as a seal
- Prevents corrosion
- Increases the efficiency of the machine

Classification of lubricants

Classification of Lubricants Based on Physical Properties

Lubricants are classified into **four main types** according to their **physical state**



Liquid Lubricants

Liquid lubricants are **fluids that reduce friction and wear between moving surfaces** by forming a thin film that separates the contacting parts. They also help in **cooling, cleaning, sealing, and protecting** machine components from corrosion.

Types of Liquid Lubricants

Animal oils: tallow oil, whale oil etc.

Vegetable oils: castor oil, palm oil etc

Petroleum oils: petroleum fractions

Synthetic lubricants: polyglycol, silicones etc

Semi-solid lubricants

Semi-solid lubricants occupy an intermediate state between liquid oils and solid greases. Commonly known as **greases**, formulated by thickening oils with soaps or non-soap thickeners.

Provide a balance between fluid mobility and structural stability.

Classification of Grease

1. Soda-based: In this case, sodium soaps are used as a thickening agent in mineral or petroleum oil. They are **slightly soluble in water**. They can be used **up to 175°C**.

2. Lithium-based: In this case, lithium soaps are emulsified with petroleum oil. They are **water-resistant** and can be **used up to 15°C**.

3. Calcium-based: In this case, calcium soaps are emulsified with petroleum oil. They are also water-resistant and can be used up to 80°C. At higher temperatures, soap and petroleum oil are separate from each other.

Solid Lubricants

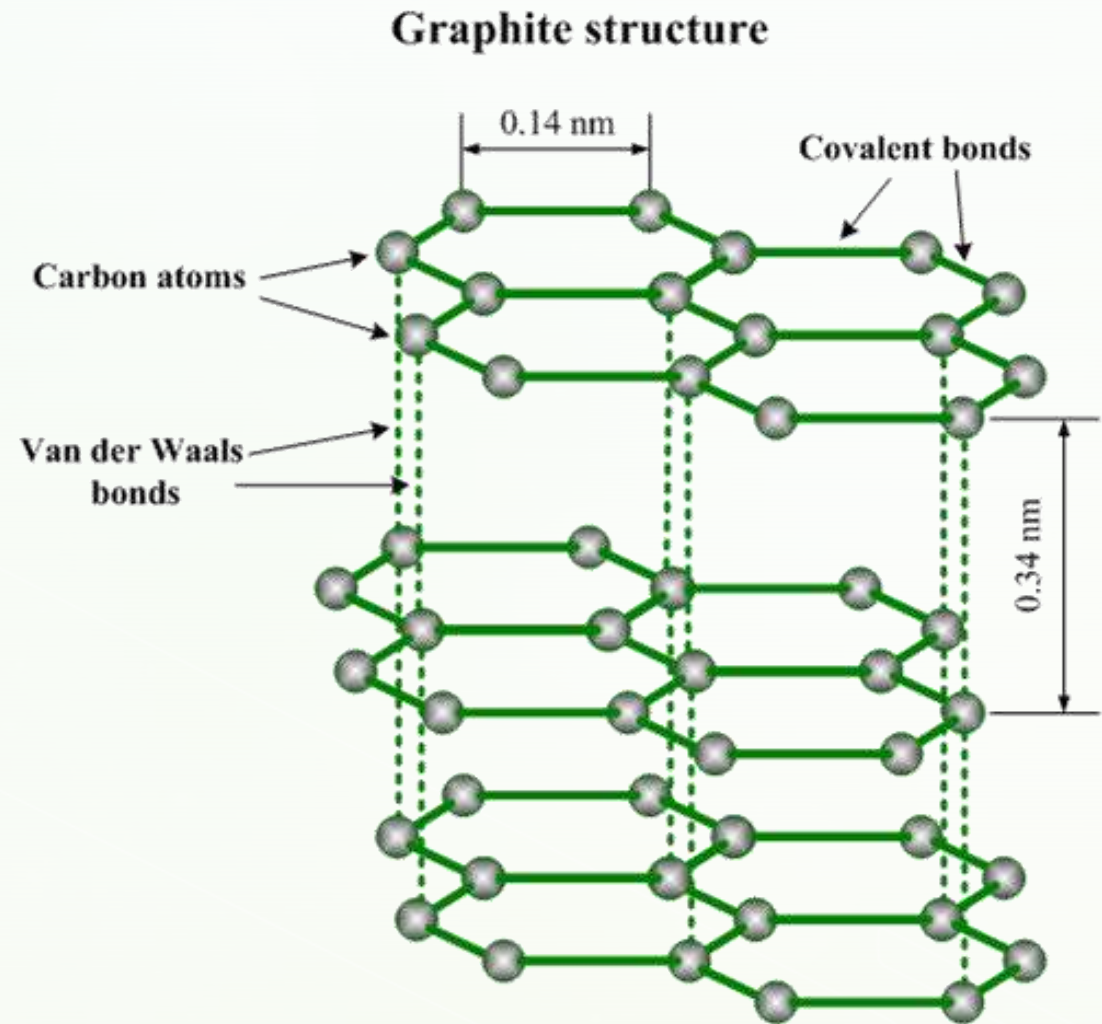
Graphite, molybdenum disulfide (MoS₂), **PTFE (Teflon)** and boron nitride (BN)_x are predominantly used as solid lubricants. They are used under high temperature and high load (pressure).

i. Graphite:

- ✓ It is most widely used as a solid lubricant.
- ✓ Graphite has a layer structure; layers are held together with the help of weak van der Waals forces, which facilitate the easy sliding of one layer on the other layer.
- ✓ It is very soapy to touch, non-inflammable.
- ✓ It is used at higher temperatures (around 450°C). They are either used in powder form or mixed with oil or water.



Graphite

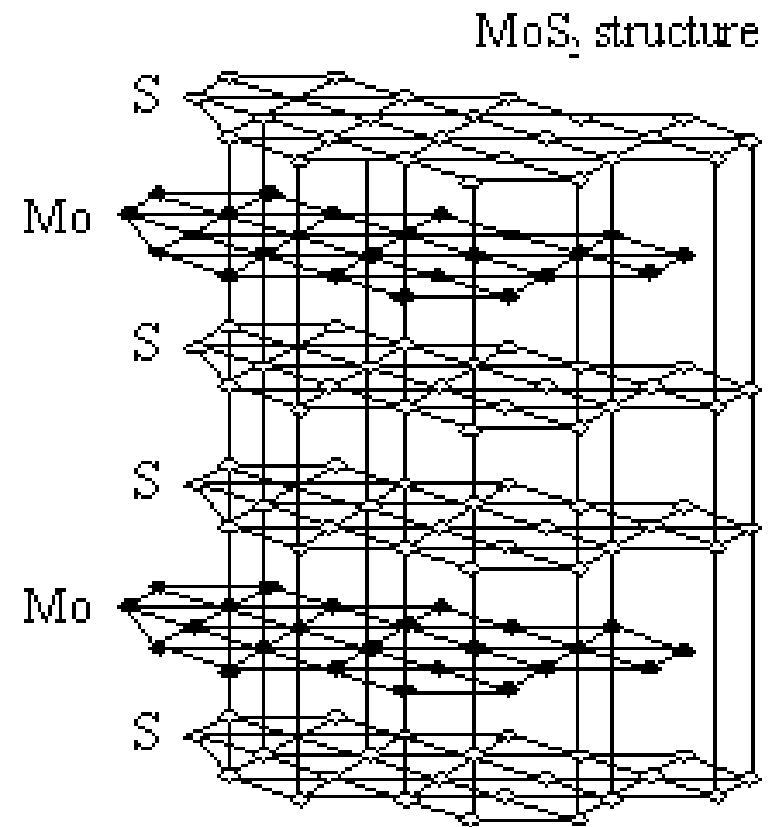


ii) Molybdenum disulphide (MoS₂):

- It is a **sandwich-like structure** in which a hexagonal layer of molybdenum (Mo) lies between two hexagonal layers of sulfur (S) atoms.
- Like graphite, each layers are held together with weak van der Waals' forces.
- It is stable up to 400°C in air and 1100 °C in vacuum.
- It differs from graphite because **it is used in high vacuum unlike graphite** (graphite is mixed with water or oil).
- It adheres even more strongly to the metal or other surface.



Molybdenum disulphide



Properties of Lubricants

- a) **Cloud Point:** When we cool down the oil, the temperature at which lubricating oil becomes **cloudy in appearance** is called the cloud point.

- b) **Pour Point:** The lowest temperature at which the lubricant oil becomes **semi-solid and ceases to flow** is called pour point. It indicates the **suitability of lubricants** used in cold conditions. A good lubricant should possess a low pour point.

c) Flash point: The flash point of a volatile material is the lowest temperature at which vapors of the material will **ignite for a moment** when an ignition source brought near to it. The lubricating oil should have flash point reasonably above its working temperature.

d) Fire point: The fire point of a fuel is the lowest temperature at which the vapour of that fuel will **continue to burn for at least 5 seconds** when an ignition source brought near to it. Fire point is **around 10°C higher than flash point**.

e) SEN:

- The time in seconds in which oil and water separates out from its emulsion to form distinct layers.
- For a good lubricating oil SEN must be low.

f) Acid value:

- Acid value is the number of milligrams of potassium hydroxide (KOH) needed to neutralize the free fatty acids in one gram of a fat or oil sample. It indicates the degree of rancidity or hydrolysis, with a higher value signifying more breakdown of the oil.
- For a good lubricating oil acid value must be low

g) Aniline Point:

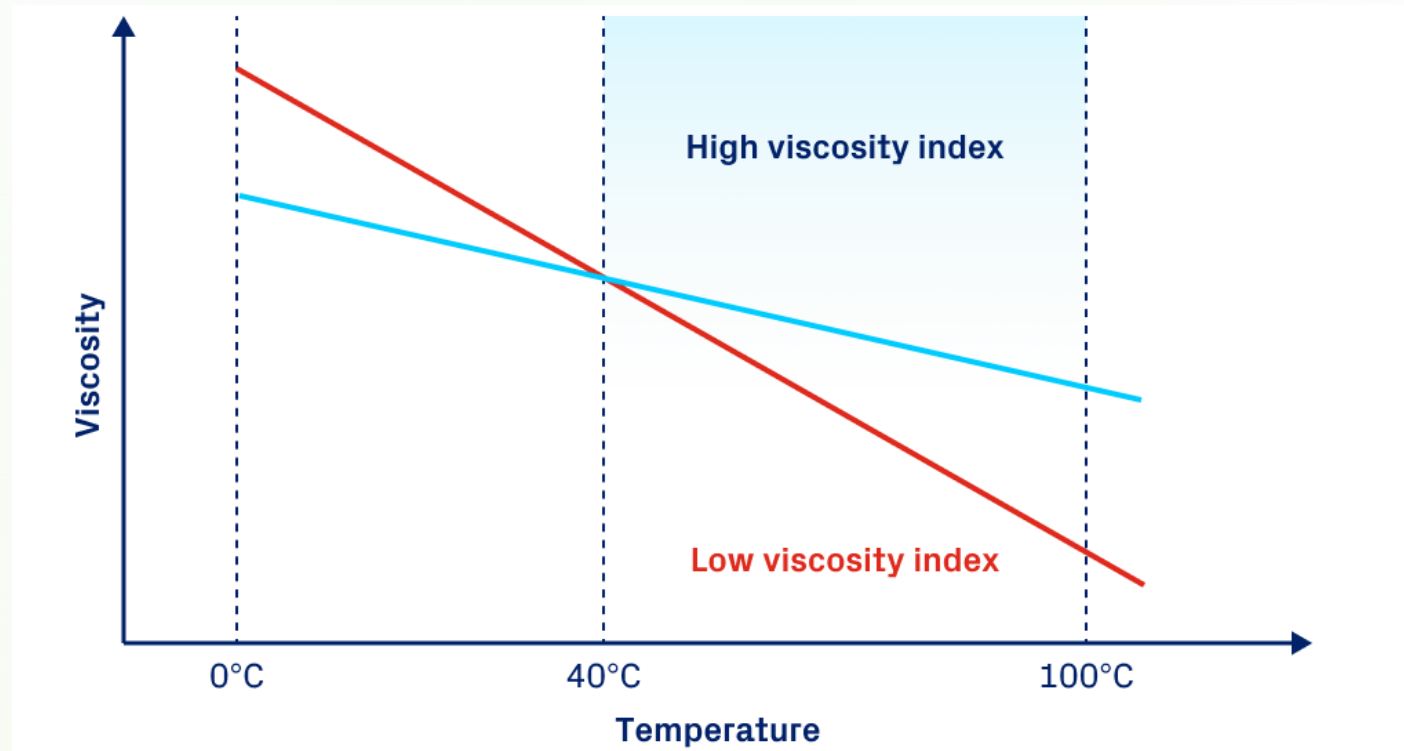
- Serves as a key indicator of an oil's aromatic content, with a higher. The aniline point is the minimum temperature at which equal volumes of aniline and a sample (like oil) become completely miscible, forming a single phase.
- Its significance is that it aniline point indicating a higher concentration of paraffinic (alkenes) hydrocarbons. This is also important for evaluating fuel quality, as it helps estimate knocking sound in the engine caused by long straight chain hydrocarbons.
- With a higher aniline point indicating a lower concentration of long straight chain hydrocarbons.

h) Viscosity:

- Viscosity is the property of a fluid that determines its **resistance to flow**.
- It is an indicator of the flow ability of lubricating oil.
- The **lower the viscosity**, the **greater the fluid's flowability**.
- If **temperature increases** viscosity of the lubricating oil **decreases** and **pressure increases** viscosity of lubricating oil **increases**.
- In short, we can say that **good lubricating oil** is that whose viscosity does not change with temperature.
- Unit of viscosity is poise or centipoise

i) Viscosity Index:

- The **variation of viscosity** of a liquid with **temperature** is called the viscosity index.
- A relatively **small change in viscosity** with temperature is indicated by a **high viscosity index**(Pennsylvania oil), whereas a **low viscosity index** shows a relatively **large change in viscosity** (Gulf oil) with temperature.



The viscosity index of the test oil is determined with the help of two types of standard oils:

1) Pennsylvanian oil (V.I. = 100)

2) Gulf oil (V.I. = 0)

$$V.I. = (L-U/L-H)*100$$

Where;

L- Viscosity of oil (Gulf oil) at $100^{\circ}F$, which having a low viscosity index

H- Viscosity of oil (Pennsylvania oil) at $100^{\circ}F$ that has a High viscosity index

U- Viscosity of sample oil at $100^{\circ}F$

Numerical on V.I.

Question 1: An oil of unknown viscosity- the index has a Saybolt Universal viscosity of 60 seconds at 120 °F and of 600 Seconds at 100 °F. The high viscosity index standard (i.e. Pennsylvanian oil has a Saybolt viscosity of 60 seconds at 210 °F and 500 seconds at 100 °F. The low viscosity-index standard (i.e. Gulf) oil has a Saybolt universal viscosity of 60 seconds at 210 °F and 800 seconds at 100 °F. Calculate the viscosity index of the unknown oil.

Sol.

Given, U = 600 s, L = 800 s, H = 500 s

$$V.I. = (L-U/L-H)*100$$

$$VI = (800-600/800-500)*100$$

$$VI = (200/300)*100$$

$$VI = 66.66$$

Question 2: An oil sample under test has a Saybolt universal viscosity of 64 seconds at 210 °F and 564 seconds at 100 °F. The low viscosity standard(Gulf oil) possesses a Saybolt viscosity of 64 seconds at 210 °F 774 seconds at 100 °F. The high viscosity index standard (i.e. Pennsylvanian oil) has a Saybolt viscosity of 60 seconds at 210 °F and 414 seconds at 100 °F. Calculate the Viscosity index of the oil sample under test.

Sol.

Given, U = 564 s, L = 774 s, H = 414 s

$$V.I. = (L-U/L-H)*100$$

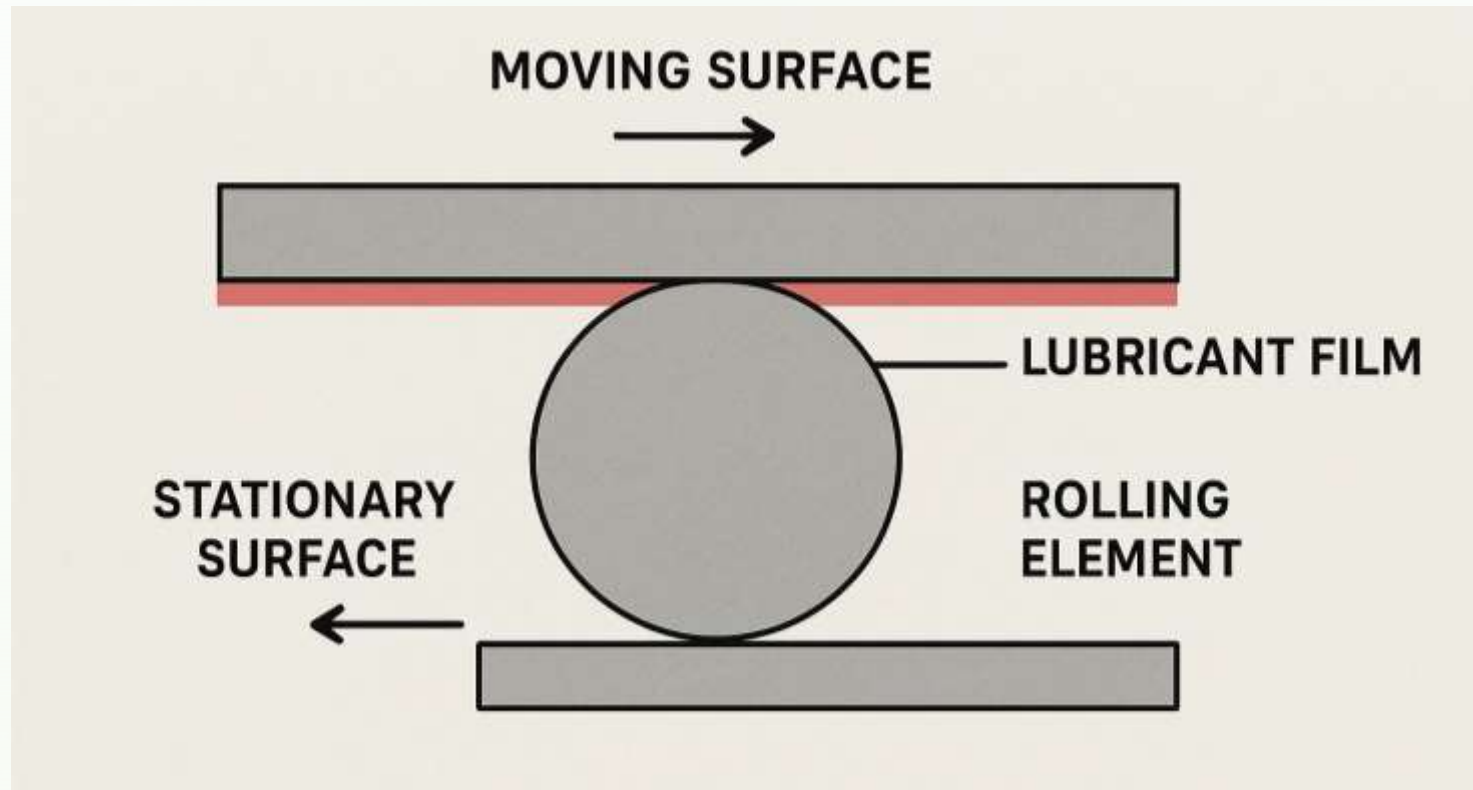
$$VI = (774-564/774-414)*100$$

$$VI = (210/360)*100$$

$$VI = 58.33$$

Mechanism of Lubrication

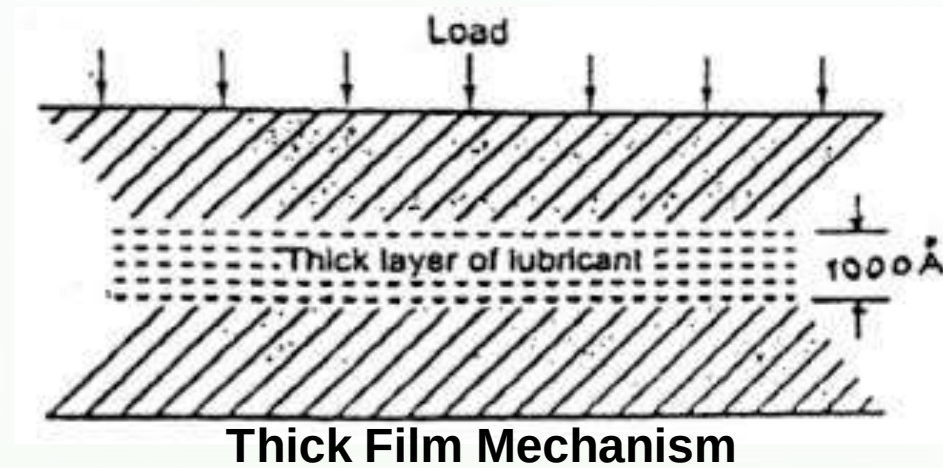
Lubrication works by **reducing friction and wear** between moving surfaces through the formation of a film (liquid, solid, or gas) that separates them.



1. Thick Film Mechanism (Hydrodynamic Lubrication)

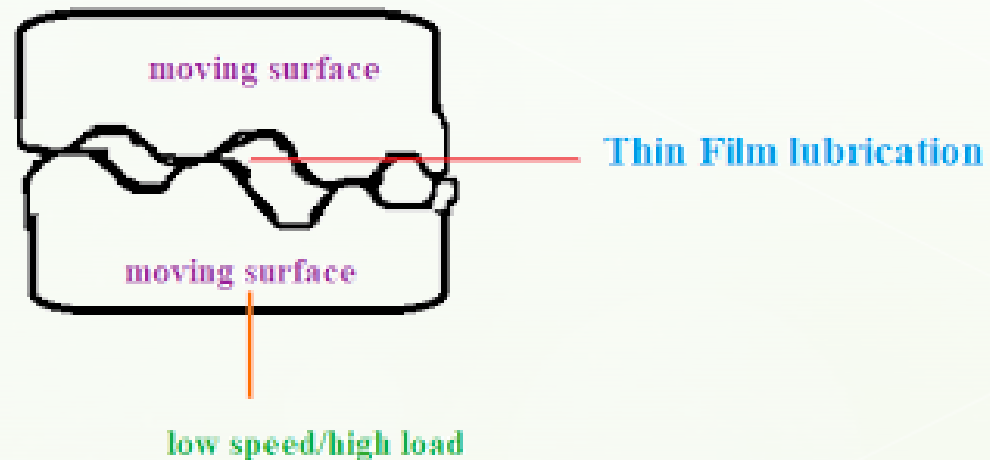
- In the **thick film mechanism**, a **complete film of lubricant** separates the two moving surfaces.
- The **lubricant film thickness (1000 Å)** is much greater than the surface roughness, so **no direct metal-to-metal contact** occurs.
- The **load** is carried entirely by the **pressure developed within the moving lubricant film**.
- This pressure is generated due to the **relative motion** of surfaces and the **viscosity** of the lubricant.
- As a result, **friction and wear are greatly reduced**, and **smooth motion** is achieved.

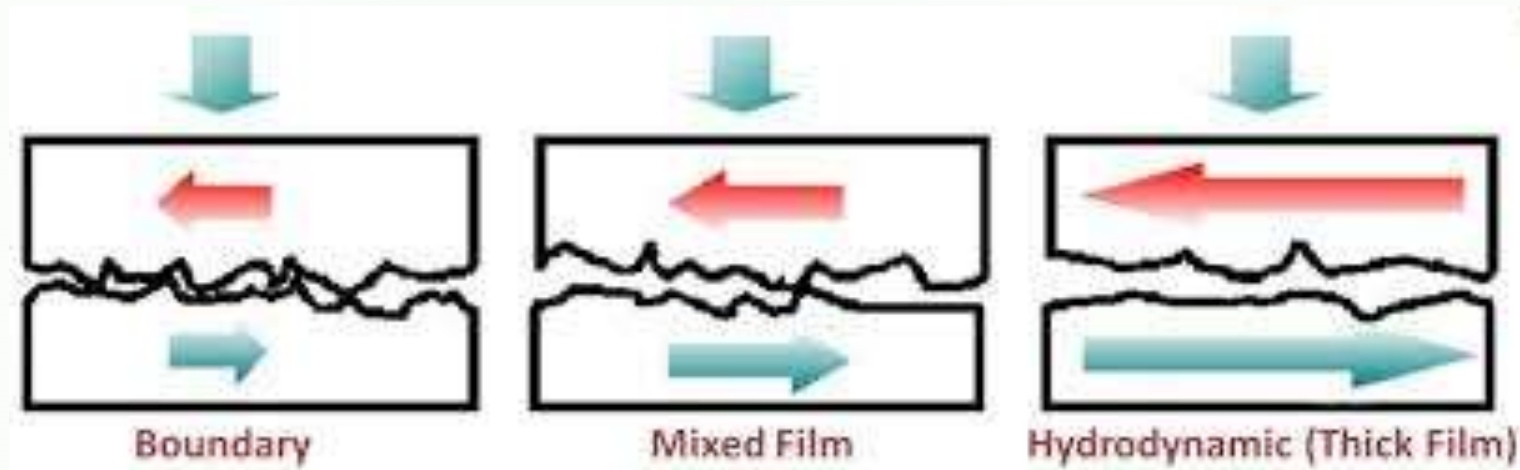
Example: Found in journal bearings, engine crankshafts, and turbines.



2. Thin Film Lubrication

Thin film lubrication (also known as **boundary lubrication**) occurs when the lubricant film is **very thin**, often just a few molecular layers thick, not enough to separate the two moving surfaces completely.



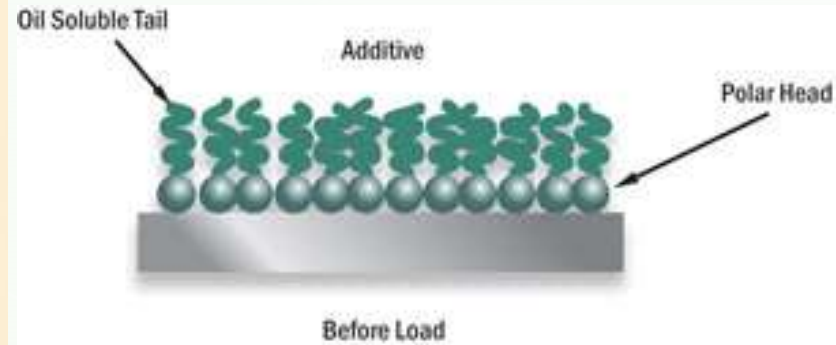
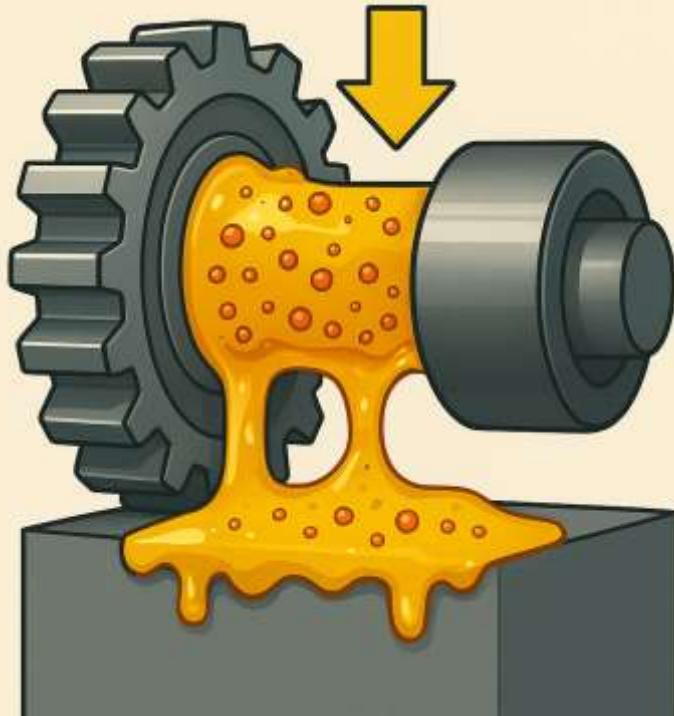


3. Extreme pressure (or temperature) lubrication

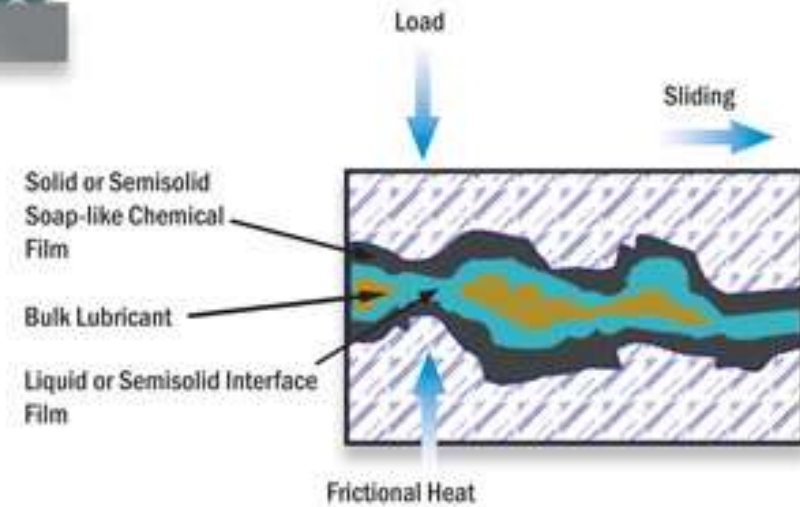
- ❑ In this mechanism, moving or sliding surface are under **high pressure** and **speed** therefore it is known as extreme pressure lubrication.
- ❑ In such a case **high temperature generated** due to friction under these condition **liquid lubricant are fail to stick and decompose or vaporize**.

- ❑ These problem are minimize by adding special additive to mineral oils. These additive form durable film on metal surface which can with stand high load and high temp.
- ❑ Important additives are organic compound groups like chloride, sulphur, phosphorus, etc.
- ❑ They react with metallic surfaces to form metallic compounds (possess high melting point, serve as good lubricant under extreme temperature and pressure), like metal chloride, Sulphide and phosphate as a more durable film.

EXTREME PRESSURE LUBRICATION



Under boundary conditions, antiwear film shears instead of surface material. However, mild chemical wear (chemical polishing) does result.



FUEL & COMBUSTION



Fuel

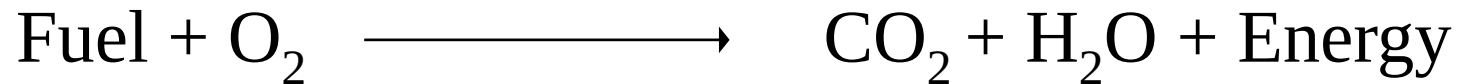


Energy



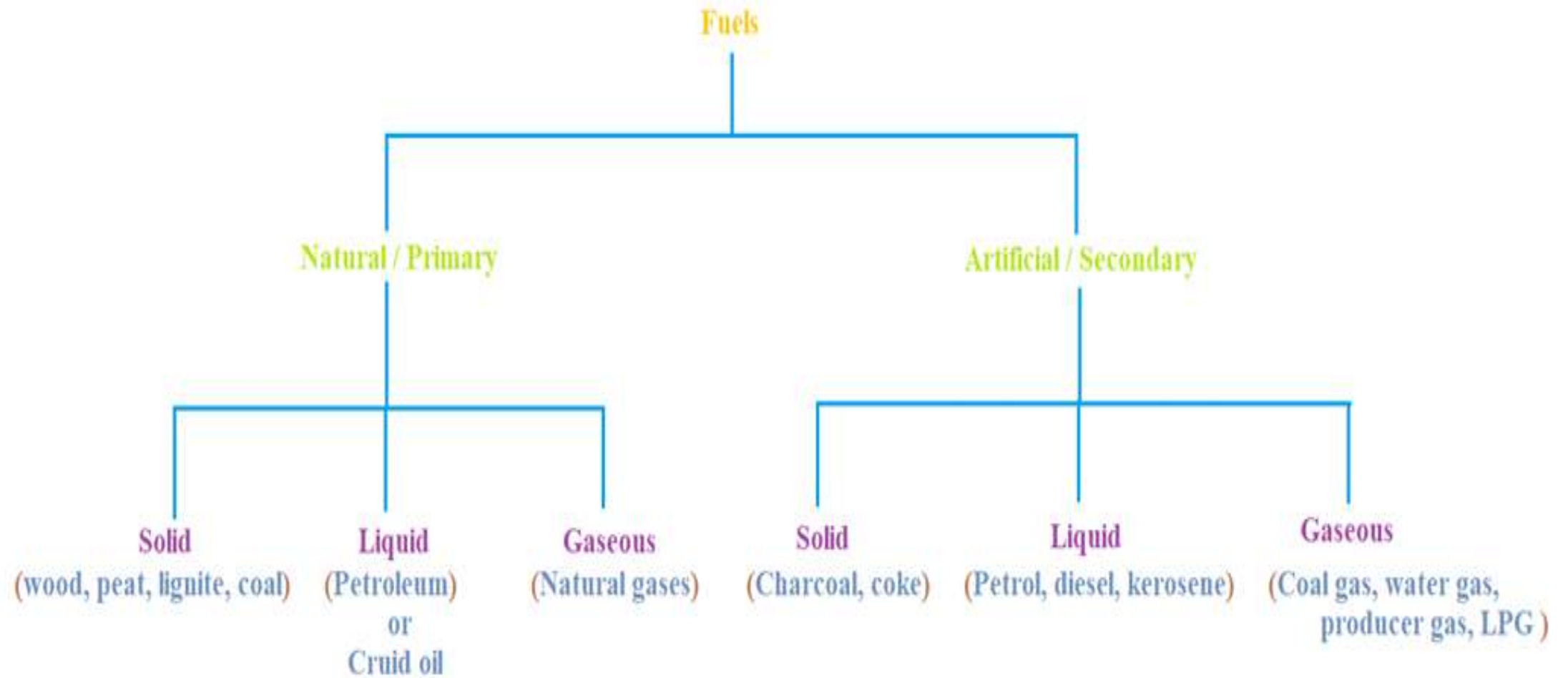
Introduction:

- The fuel is a material that, when raised to its ignition temperature, continues to burn if sufficient oxygen or air is available.
- Fuel is a combustible substance containing carbon as the main constituent, which, on proper burning gives a large amount of heat that can be used economically for domestic and industrial purposes.



- The various fuels used economically are wood, coal, kerosene, petrol, diesel, gasoline, coal gas, producer gas, water gas, natural gas, etc.

Classification of Fuels



Characteristics of Good Fuel:

- Fuel should have a high calorific value.
- Must have moderate ignition temperature.
- Fuel should have low moisture content.
- Available in bulk at low cost.
- Should not burn spontaneously.
- Fuel should burn efficiently, without releasing hazardous pollutants.
- Handling, storage, and transportation should be easy.

Calorific value

- The calorific value of fuel can be defined as the amount of heat evolved when one unit mass or volume of the fuel undergoes complete combustion in the presence of oxygen.
- **High or gross calorific value (HCV or GCV):** It is defined as the amount of heat evolved when one unit mass or volume of the fuel is completely burnt and combustible products are cooled to room temperature (25 °C or 77°F).
- **Low or net calorific value (LCV or NCV):** It is defined as the amount of heat evolved when one unit mass or volume of the fuel is completely burnt and combustible products are permitted to escape. Therefore, net calorific value is lower than the gross calorific value.
- $LCV = HCV - \text{latent heat of water vapor}$
- $LCV = HCV - \text{mass of hydrogen} \times 9 \times \text{latent heat of steam (587 kcal/kg)}$
- One part by mass of hydrogen produces nine parts by mass of water molecule. Therefore, $LCV = HCV - H/100 \times 9 \times 587 \text{ kcal/kg}$
- H = percentage of hydrogen in fuel

Units of Heat

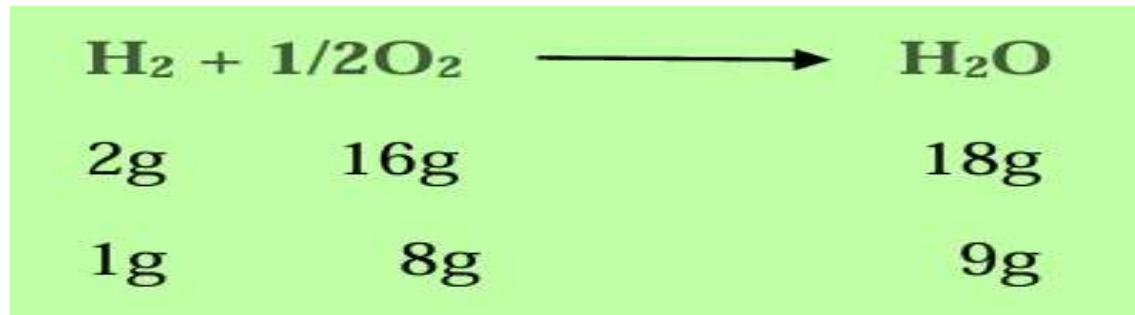
- A **calorie** is the amount of heat required to raise the temperature of one gram of water through one degree centigrade (15-16 °C).
- A **kilocalorie** is equal to 1,000 calories. This is the unit of the metric system and may be defined as “the quantity of heat required to raise the temperature of one kilogram of water through one degree centigrade. Thus, 1 kcal = 1,000 calories.

Theoretical calculation of Calorific value by Dulong's formula

- The calorific value of fuel can be approximately computed by noting the amounts of the constituents of the fuel. The higher calorific value of some of the chief combustible constituents of fuel is tabulated below:

Constituent	Carbon	Hydrogen	Sulphur
HCV (kcal/kg)	8,080	34500	2,240

- The oxygen, if present in the fuel, is assumed to be present in combined form with hydrogen, i.e., in the form of fixed hydrogen (H_2O). So, the amount of hydrogen available for combustion = Total mass of hydrogen in fuel – Fixed hydrogen



Total mass of hydrogen in fuel – fixed hydrogen

Total mass of hydrogen in fuel – (1/8) mass of oxygen in the fuel

$\therefore$ 8 part of oxygen combine with 1 part of hydrogen to form H_2O .

$$\text{Fixed hydrogen} = \frac{\text{Mass of oxygen in the fuel}}{8}$$

Then the amount of hydrogen available for combustion = total mass of hydrogen in fuel – fixed hydrogen = $[\text{H}-\text{O}/8]$

Dulong's formula

$$\text{HCV} = \frac{1}{100} [8080 C + 34500 \left(H - \frac{O}{8} \right) + 2240 S] \text{ kcal/kg}$$

$$\text{LCV} = \text{HCV} - \frac{H}{100} \times 9 \times 587 \text{ kcal/kg}$$

Numericals based on Dulong's Formula

- **Problem 1:** Calculate the gross and net calorific values of a coal sample having 85 % C, 8 %H, 1 % S, 2% N and 4% ash. Latent heat of steam = 587 kcal/kg.
- **Problem 2:** A sample of coal has the following composition: C = 82 %, H= 7.5 %, S = 1.5 %, N = 0.6 %, O = 8.4 %. Find GCV using Dulong's Formula.
- **Problem 3:** Calculate GCV and NCV of coal sample having the following composition: C = 80%, H = 7%, S= 3.5 %, N = 2.1%, ash = 4.4%.

- 1. (Ans: GCV=9650.4 kcal/kg and NCV= 9227.76 kcal/kg)
- 2. (Ans: GCV=8965.25 kcal/kg)
- 3. (Ans: GCV=8828.02 kcal/kg and NCV= 8458.2 kcal/kg)

Solution1

- **Given:** C = 85%, H = 8%, S = 1%, N = 2%, Ash = 4%
- %of oxygen= O=100-(C+H+S+N+Ash)=100-100=0%

Step 1: Substitute in Dulong's formula

- $$HCV = \frac{1}{100} [8080(85) + 34500(8 - 0/8) + 2240(1)]$$

- $$HCV = \frac{1}{100} [8080 \times 85 + 34500 \times 8 + 2240 \times 1]$$

• Step 2: Simplify each term

- $8080 \times 85 = 686800$

- $34500 \times 8 = 276000$

- $2240 \times 1 = 2240$

- Add them together:

- $686800 + 276000 + 2240 = 965040$

- $$HCV = \frac{965040}{100} = \mathbf{9650.4 \text{ kcal/kg}}$$

• Step 3: Calculate Net Calorific Value (NCV)

- $$NCV = HCV - \left(\frac{9H}{100} \right) \times L \quad (\text{Where } L = 587 \text{ kcal/kg})$$

- $$NCV = 9650.4 - \left(\frac{9 \times 8}{100} \right) \times 587, \quad NCV = 9650.4 - 0.72 \times 587$$

- $$NCV = 9650.4 - 422.64 = \mathbf{9227.76 \text{ kcal/kg}}$$

- **Solution2**
- **Formula (Dulong's Formula for Gross Calorific Value, GCV / HCV):**
- $$\text{GCV} = \frac{1}{100} \left[8080 C + 34500 \left(H - \frac{O}{8} \right) + 2240 S \right]$$

Step 1: Substitute the given values

- $$\text{GCV} = \frac{1}{100} \left[8080(82) + 34500 \left(7.5 - \frac{8.4}{8} \right) + 2240(1.5) \right]$$

- **Step 2: Simplify the hydrogen term**

- $$\frac{O}{8} = \frac{8.4}{8} = 1.05, \quad H - \frac{O}{8} = 7.5 - 1.05 = 6.45$$

- **Step 3: Substitute values into the main equation**

- $$\text{GCV} = \frac{1}{100} [8080(82) + 34500(6.45) + 2240(1.5)]$$

- **Step 4: Calculate each term**

- $$8080 \times 82 = 662560$$

- $$34500 \times 6.45 = 222525$$

- $$2240 \times 1.5 = 3360$$

- $$662560 + 222525 + 3360 = 888445$$

- **Step 5: Divide by 100**

- $$\text{GCV} = \frac{888445}{100} = \mathbf{8884.45 \text{ kcal/kg}^{**}}$$

- **Solution 3: Find Oxygen (O) percentage**

- $O = 100 - (C + H + S + N + Ash)$
 - $O = 100 - (80 + 7 + 3.5 + 2.1 + 4.4) = 100 - 97 = 3\%$
 - $GCV = \frac{1}{100} \left[8080C + 34500 \left(H - \frac{O}{8} \right) + 2240S \right]$
 - $GCV = \frac{1}{100} \left[8080(80) + 34500 \left(7 - \frac{3}{8} \right) + 2240(3.5) \right]$
 - $\frac{O}{8} = \frac{3}{8} = 0.375$
 - $H - \frac{O}{8} = 7 - 0.375 = 6.625$
 - $8080 \times 80 = 646400$
 - $34500 \times 6.625 = 228562.5$
 - $2240 \times 3.5 = 7840$
 - $646400 + 228562.5 + 7840 = 882802.5$
 - $GCV = \frac{882802.5}{100} = \mathbf{8828.03 \text{ kcal/kg}}$
- $$NCV = GCV - \left(\frac{9H}{100} \right) \times L, \quad NCV = 8828.03 - \left(\frac{9 \times 7}{100} \right) \times 587$$
- $NCV = 8828.03 - (0.63 \times 587)$
 - $NCV = 8828.03 - 369.81 = \mathbf{8458.22 \text{ kcal/kg}}$